# XRPL Validator — Daily Engineering Report · 2026-10-16

*Period: 2026-10-16 00:00 to 24:00 US Eastern (2026-10-16 04:00 to 2026-10-17 04:00 UTC); ledgers #107,687,073 to #107,709,339. Every count from the ledger and the validator's logs stops at #107,709,339, closed 03:59:51 UTC. Generated 2026-10-21 00:48 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.*

*Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records; Step 1 receipt records (they begin 2026-08-29); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); xrpl-validator merge records (they begin 2026-07-04); amendment data; data.xrpl.org agreement scores. Not available: none.*

## Status

**Impaired** (all causes resolved or fixed) — 8 validations of a ledger hash the network did not validate.

*Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.*

## Executive summary

- **Validator:** signed 22,250 of 22,267 ledgers; 8 validations did NOT match the network's validated ledger. Unsigned: 17 during planned restarts.
- **Native engine:** 2 engine receipts recorded (2 fixed, 0 explained, 0 without a recorded fix).
- **Changes:** 2 deployments (cycles 180, 181), 2 commits merged into main, 1 engine finding fixed (F421).
- **Network:** 1 slow-close period; the longest, 15:58 UTC–15:59 UTC, closed ledgers up to 19 s apart.
- **Amendment fixCleanup3_4_0** holds majority; activation due ~2026-10-23 09:34 UTC.

## Service levels

| Measure | Value |
|---|---|
| Ledgers closed by the network | 22,267 |
| Validations issued | 22,250 |
| Coverage (validations / ledgers closed) | 99.924% |
| Validations matching the network's validated ledger hash | 22,242 of 22,250 |
| Unsigned during planned restarts | 17 |
| Unsigned outside maintenance | 0 |

| Measure | Value |
|---|---|
| Coverage outside planned maintenance (objective 99.9%) | 100.000% |
| Restarts | 2 |
| Basis: the hash the first peer announced as accepted (before cycle 183) | 22,250 |
| External agreement, data.xrpl.org (UTC day 2026-10-16) | 0.99805 (38 missed of 19,467) |

Restarts:

- 16:58:23 UTC: deployment of cycle 180; validations resumed at #107,699,117 (9 ledgers unsigned between the last validation before the restart and the first after it)
- 19:38:36 UTC: deployment of cycle 181; validations resumed at #107,701,566 (8 ledgers unsigned between the last validation before the restart and the first after it)

*The basis rows sum to validations issued. Coverage counts the validations the validator issued for ledgers the network closed in the period. Agreement compares each validation's ledger hash with the network's validated hash for that ledger. data.xrpl.org scores UTC days from the validations its crawlers received, so it can differ.*

## Native engine conformance

Ledgers with a receipt in the period (#107,687,073–#107,709,339), by step:

| Step | Ledgers with a receipt |
|---|---|
| Step 1 · native engine | 1 |
| Step 2 · records | 1 |
| Step 3 · order | 1 |
| Step 4 · peer data | 0 |
| Step 4 · validations | 8 |
| Step 5 · own hash | 1 |
| Writer | 1 |

Live counters are kept only for the current run; none exist for a past day.

## Receipt register

| ID | Time | Ledger(s) | Step | Description | Disposition |
|---|---|---|---|---|---|
| R-01 | 13:35 UTC | #107,696,010 | Step 1 · native engine, Step 2 · records, Step 5 · own hash, Writer | OfferCreate tx 548165639AD7: Offer 963752A2 (created on the network): NewFields.TakerPays ours="99997497175367033" theirs="999974… | fixed: F421: a sell offer's leftover is repriced through the whole-drop rate as rippled does, live since 10-16 12:59 (cycle 180) |
| R-02 | 18:43 UTC | #107,700,727–#107,700,734 (8) | Step 4 · validations | 27 of the 28 trusted validations needed reached the validator in time | explained: 27 of the 28 trusted validations needed reached us in time (some came late); the network validated the ledger normally (all signed) |
| R-03 | 18:43 UTC | #107,700,728 | Step 3 · order | our transaction order differed at position 0; 4 result(s) differed; 57 object(s) differed | fixed: Step 3 takes the set more than half the trusted validators name, so a few late validations no |

| ID | Time | Ledger(s) | Step | Description | Disposition |
|---|---|---|---|---|---|
| | | | | | longer change the order, live since 10-16 15:39 (cycle 181) |

*A receipt is recorded whenever the validator's own work differs from the network's, or the network's trusted validations reached it short of quorum. One ledger flagged by several steps is one receipt. Dispositions: fixed (with the finding, commit and deployment), explained (not a defect, with the reason), or open. Receipt and incident IDs are numbered per report: R-01 here is R-2026-10-16-01, INC-01 is INC-2026-10-16-01.*

## Incidents

### INC-01 · 8 validations of a ledger hash the network did not validate

| | |
|---|---|
| Window | 2026-10-16 06:28 UTC to 2026-10-17 00:59 UTC |
| Ledgers | #107,689,381, #107,691,673, #107,697,983, #107,698,197, #107,702,560, #107,704,140, #107,705,574, #107,706,546 |
| Impact | the validator signed a hash the network did not validate for these ledgers; data.xrpl.org scores such ledgers missed |
| Cause | Before cycle 183 the validator signed the ledger hash from the first peer that announced a ledger as accepted, without confirmation. For these ledgers that peer had built a different ledger, so the validator signed a hash the network never validated. |
| Action | commit d540e6e — fix(validator): sign a ledger only once the trusted validators reach quorum on its hash; deployed in cycle 183 at 2026-10-18 22:18 UTC |

## Change log

### Deployments

| Cycle | Main | Deployed | Test suites | Replay windows | Post-deploy health | Soak |
|---|---|---|---|---|---|---|
| 180 | `fbcc3ae` | 16:59:13 UTC | 2,157 passed / 0 failed; port 1,006 / 0 | HIST60 60 ledgers, 0 mismatches, WIN98 98 ledgers, 0 mismatches | 404 matched, 0 mismatched | #55 from 17:25 UTC: ended early: superseded by the next deployment (2026-10-16 19:39 UTC) |
| 181 | `b6948f4` | 19:39:26 UTC | 2,157 passed / 0 failed; port 1,006 / 0 | HIST60 60 ledgers, 0 mismatches, WIN98 98 ledgers, 0 mismatches | 404 matched, 0 mismatched | #56 from 20:05 UTC: CLEAN (2026-10-17 20:12 UTC) |

Cycle 180 delivered: `1711cca` chore(tools): xrpl-watch moved to its own repo (ilubxrp589/xrpl-watch); this folder keeps a pointer; `0bb7783` chore: scrub a login password and home-network addresses from docs, the dashboard config and a script; `67ce8ff` chore(node): the source names no home-network address; `fbcc3ae` fix(ledger): F421 — a crossed offer's remainder is repriced through the native rate (a 17-digit TakerPays is not shrunk)

Cycle 181 delivered: `b6948f4` fix(node): Step 3 salts with the set more than half the trusted list names, not only a full quorum

*Each deployment passes the unit and port test suites and replays mainnet history windows with zero mismatches before it ships, is health-checked after it starts, and then soaks for 24 hours.*

### Merged into main

| Merged | Repo | Commit | Subject |
|---|---|---|---|
| 13:47 UTC | xrpl-validator | `fbcc3ae` | fix(ledger): F421 — a crossed offer's remainder is repriced through the native rate (a 17-digit TakerPays is not shrunk) |
| 18:55 UTC | xrpl-validator | `b6948f4` | fix(node): Step 3 salts with the set more than half the trusted list names, not only a full quorum |

## Findings fixed

- **fix(ledger): F421 — a crossed offer's remainder is repriced through the native rate (a 17-digit TakerPays is not shrunk)** ( `fbcc3ae` , xrpl-validator) flowCross reprices the rested remainder through Quality{takerAmount.out, takerAmount.in}.rate(), and rippled's divide never shrinks a native mantissa (Finding 240): with a seventeen-digit drop count the quotient is sixteen digits and the +5 lands on its last digit. The book page already used keylet::rate_encode_native; the three reprice sites in the placement block still used rate_of_me (the IOU rule), so the …

### Other engineering changes

- **fix(node): Step 3 salts with the set more than half the trusted list names, not only a full quorum** ( `b6948f4` , xrpl-validator) Mainnet #107700728 (2026-10-16, soak #55): only 27 trusted validations of quorum 28 reached us on #107700727-734, so agreed_set gave None and Step 3 salted with the ledger's own root (salt_own=8). On #107700728 the consensus set held a transaction the ledger never filed, so our order differed at position 0 (382 = 382 transactions, 4 results and 57 objects differed). The 27 that did reach us all named the network's …

## Independence program

No independence record for UTC day 2026-10-16; records begin 2026-10-20.

## Network

| Measure | Value |
|---|---|
| Ledgers closed | 22,267 (#107,687,073–#107,709,339) |
| Mean close interval | 3.88 s |
| Longest close interval | 19 s (before #107,698,198) |
| Closes at least 15 s apart | 1 |
| Empty ledgers | 0 |

- Slow closes 15:58:51 UTC–15:59:10 UTC: ledgers #107,698,198, 1 close at least 15 s apart, up to 19 s.

*From ledger close times, which the network records at a 10-second resolution or coarser, so single intervals are approximate.*

## Amendment watch

*Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.*

| Amendment | Majority since | Activation due |
|---|---|---|
| fixCleanup3_4_0 | 2026-10-09 09:34 UTC | 2026-10-23 09:34 UTC |

Amendments enabled on mainnet: 97.

## Open items

- Amendment fixCleanup3_4_0: activation due ~2026-10-23 09:34 UTC.

## Corrections to earlier reports

None.

## Appendix — commits

| Committed | Merged into main | Repo | Commit | Subject |
|---|---|---|---|---|
| 10-16 13:43 | 10-16 13:47 | xrpl-validator | `fbcc3ae` | fix(ledger): F421 — a crossed offer's remainder is repriced through the native rate (a 17-digit TakerPays is not shrunk) |
| 10-16 18:54 | 10-16 18:55 | xrpl-validator | `b6948f4` | fix(node): Step 3 salts with the set more than half the trusted list names, not only a full quorum |

*Times in UTC. A commit appears on the day it was committed to main or merged into it.*